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Reinforcement-Learning-Based Hyper-Heuristic for Feature Selection in Disease Prediction

Emerging Topics in AI — Option 2: Real-World AI Research and Implementation Project

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Abstract

Feature selection can reduce model complexity and improve interpretability in healthcare prediction, but the usefulness of a fixed selection strategy may vary during search. This study investigates a Q-learning-based hyper-heuristic that adaptively chooses among four low-level feature-selection operators—ADD, REMOVE, SWAP, and FLIP—rather than selecting features through a single fixed rule. Logistic Regression is used to evaluate candidate subsets, and the optimization objective balances ROC-AUC with subset compactness. The primary evaluation uses the CDC Diabetes Health Indicators dataset with 253,680 observations and 21 predictors, while the UCI Cleveland Heart Disease dataset with 303 observations and 13 predictors provides a secondary validation setting. The proposed method is compared with all-feature models, random and fixed heuristic strategies, SelectKBest, L1-based selection, and Recursive Feature Elimination (RFE). Repeated experiments are assessed using paired Wilcoxon signed-rank tests with Holm correction. On Diabetes, RL achieved a mean ROC-AUC of 0.8210 using 13.8 features on average, corresponding to a 34.3% reduction, while preserving performance close to the all-feature AUC of 0.8230. However, RFE and L1 achieved stronger overall performance-compactness trade-offs. On Heart Disease, RL reduced the feature set by 38.2% but achieved a lower and more variable mean ROC-AUC of 0.8791 compared with 0.9032 for all features. The results show that reinforcement-learning hyper-heuristics are viable adaptive search mechanisms for feature selection, but their advantage depends on dataset scale, validation stability, and the strength of conventional baselines.

Index Terms—reinforcement learning, Q-learning, hyper-heuristics, feature selection, healthcare AI, disease prediction, diabetes, heart disease, logistic regression.

I. INTRODUCTION

Healthcare machine-learning systems frequently operate on datasets containing variables with unequal predictive value. Some variables may be redundant, weakly informative, or unnecessary for a particular prediction task. Feature selection seeks to retain a smaller subset of informative variables while preserving predictive performance. In healthcare applications, this can support simpler models, lower data requirements, faster computation, and potentially more interpretable decision support. These benefits motivate feature selection as both a predictive and an optimization problem [1], [2].

Traditional feature-selection methods are commonly grouped into filter, wrapper, and embedded approaches. Filter methods rank variables independently of a specific predictive model, wrappers evaluate candidate subsets through model performance, and embedded methods perform selection as part of model fitting [1], [2]. Each approach has strengths, but no single search operator or selection mechanism is expected to be best across all stages of a complex search. Hyper-heuristics address this issue at a higher level by selecting or generating heuristics instead of searching only at the direct solution level [3].

Reinforcement learning (RL) provides a natural mechanism for adaptive heuristic selection. Rather than committing to a single operator, an RL controller can observe a coarse representation of the current search condition, select a low-level heuristic, receive feedback from the resulting candidate solution, and update its heuristic-selection policy. Q-learning is particularly attractive for a compact discrete controller because it learns action values from reward feedback without requiring an explicit transition model [4]. Recent research has demonstrated the value of Q-learning for selection hyper-heuristics [5] and the broader potential of RL-driven feature selection in healthcare [6]–[8].

Motivated by these ideas, this project studies whether a Q-learning hyper-heuristic can discover compact health-related feature subsets while preserving disease-prediction performance. The CDC Diabetes Health Indicators dataset is used as the primary case study, and the UCI Cleveland Heart Disease dataset is used as a secondary validation setting. The project follows the submitted proposal in treating several outcomes as valid: the RL method may improve performance, achieve comparable performance with fewer features, or reveal that simpler baselines remain more competitive.

The main contributions of this work are as follows:

1. A Q-learning selection hyper-heuristic is designed to choose adaptively among four low-level feature-selection operators: ADD, REMOVE, SWAP, and FLIP.
2. A performance-compactness objective combines ROC-AUC with a mild feature-count penalty, allowing predictive quality and subset size to be optimized jointly.
3. The approach is evaluated on two healthcare datasets with very different sample sizes, enabling analysis of search stability across a large and a small dataset.
4. The RL controller is compared not only with random and fixed heuristic-selection strategies, but also with established feature-selection baselines: SelectKBest, L1 selection, and RFE.
5. Repeated experiments are analyzed with paired Wilcoxon signed-rank tests and Holm multiple-comparison correction to distinguish small numerical differences from statistically supported effects.
6. Cross-dataset analysis is used to identify where adaptive hyper-heuristic search is competitive and where simpler feature-selection methods remain preferable.

The remainder of the report is organized as follows. Section II reviews relevant literature. Section III formalizes the feature-selection problem and datasets. Section IV presents the proposed RL hyper-heuristic. Section V describes the experimental methodology and baselines. Section VI reports results and statistical comparisons. Section VII discusses implications and limitations, and Section VIII concludes the study and identifies future directions.

II. RELATED WORK

A. Feature Selection and Healthcare Prediction

Feature selection is motivated by predictive efficiency, cost reduction, and improved understanding of the variables that drive a learning task [1]. Wrapper methods explicitly account for the interaction between the selected subset and the learning algorithm by evaluating candidate subsets with the target classifier [2]. This model-aware property is useful in the present project because the objective is not merely to rank variables but to find subsets that retain Logistic Regression discrimination while reducing the number of predictors.

Healthcare prediction adds practical motivation for compact feature sets because each variable may correspond to a survey question, measurement, clinical test, or patient attribute. Adaptive feature acquisition and selection have therefore attracted interest in medical machine learning. Shaham et al. used reinforcement learning to select which medical-questionnaire features to reveal before making a prediction [6]. More recent studies have used multi-agent reinforcement learning for feature selection in adverse glycemic event prediction and post-operative outcome prediction, demonstrating that RL can be integrated with clinically motivated rewards and interpretability mechanisms [7], [8].

B. Hyper-Heuristics and Reinforcement Learning

Hyper-heuristics are search or learning mechanisms that operate over a space of heuristics or heuristic components rather than directly solving the underlying problem with one handcrafted algorithm [3]. In selection hyper-heuristics, a high-level method chooses among a set of low-level heuristics. This separation allows different search behaviors to be combined within one adaptive framework.

Q-learning is a model-free RL algorithm that iteratively updates estimates of the value of choosing an action in a state [4]. Zhao et al. proposed a hyper-heuristic selection using Q-learning as the high-level strategy and demonstrated that learned low-level-heuristic selection can improve search behavior on continuous optimization benchmarks [5]. Earlier work also showed that reinforcement learning can automate the design of a hyper-heuristic controller across problem domains. Separately, hyper-heuristic feature-selection research has demonstrated that composing multiple search components can provide a useful alternative to a single feature-selection method [9].

C. Research Gap and Position of This Study

The literature shows three relevant strands: conventional feature selection, hyper-heuristic optimization, and RL-based feature selection. However, the present project focuses on a specific intersection: Q-learning is used as a high-level controller that selects feature-subset modification heuristics, while a conventional classifier evaluates the resulting subsets. The goal is not to claim that RL must outperform established feature selectors, but to test whether adaptive operator selection offers a competitive and compact solution under rigorous repeated evaluation.

Table 1. Comparison of selected related work and the present study.

Reference	Domain	High-Level Method	Feature Search	Healthcare	Main Focus
Guyon & Elisseeff [1]	General ML	—	Feature-selection principles	No	FS objectives and methodology
Ibrahim et al. [9]	Benchmark datasets	Hyper-heuristic	Yes	No	Hyper-heuristic feature selection
Zhao et al. [5]	Continuous optimization	Q-learning HH	Indirect	No	Adaptive LLH selection
Shaham et al. [6]	Medical questionnaire	RL	Direct feature acquisition	Yes	Adaptive iterative feature selection
Kim et al. [7] / Woo et al. [8]	Clinical prediction	Multi-agent RL	Direct / reward-driven	Yes	RL feature selection in healthcare
This work	Diabetes + heart disease	Q-learning hyper-heuristic	ADD/REMOVE/SWAP/FLIP	Yes	Adaptive LLH selection + compact prediction

III. PROBLEM DEFINITION AND DATASETS

A. Feature-Selection Problem

Let D = {(x_i, y_i)} denote a labeled dataset with d candidate input features. A candidate feature subset is represented by a binary vector z ∈ {0,1}^d, where z_j = 1 means that feature j is selected and z_j = 0 means that it is excluded. The number of selected features is k = Σ_j z_j. The direct search spaces contain 2^21 = 2,097,152 subsets for Diabetes and 2^13 = 8,192 subsets for heart disease.

For a candidate subset, Logistic Regression is fitted on the selected variables and evaluated using ROC-AUC. The optimization objective used throughout the principal experiments is:

$$\mathbf{J(z) = ROC\text{-}AUC(z) - \lambda \times (k / d),\ \lambda = 0.02}$$

This objective assigns primary importance to predictive discrimination while applying a small penalty to large subsets. The penalty is intentionally mild: a candidate must retain strong AUC to be favored, but among similarly predictive subsets, a smaller one obtains a better score.

B. CDC Diabetes Health Indicators

The primary dataset contains 253,680 observations, 21 predictor variables, and a binary diabetes target. The selected project dataset contains no missing values. The target is imbalanced: approximately 86.07% of observations belong to the non-diabetes class and 13.93% to the diabetes class. Because of this imbalance, Diabetes Logistic Regression models use balanced class weighting. The source survey is based on the CDC Behavioral Risk Factor Surveillance System (BRFSS), a large telephone-based health survey program [10].

C. UCI Cleveland Heart Disease

The secondary dataset is the processed Cleveland subset of the UCI Heart Disease database. It contains 303 observations and 13 input features. The original target ranges from 0 to 4; the project converts it into a binary endpoint where values greater than zero indicate presence of heart disease. Six missing values are present in the processed data (four in ca and two in thal) and are handled by median imputation. UCI documents the Cleveland database as a 303-instance, 13-feature classification dataset commonly evaluated by separating target value 0 from values 1–4 [11].

Table 2. Dataset characteristics used in the experiments.

Characteristic	Diabetes	Heart Disease
Observations	253,680	303
Input features	21	13
Target	Binary diabetes	Binary heart disease (num > 0)
Missing values	0	6
Class handling	Balanced class weights	Approximately balanced
Preprocessing	Standardization	Median imputation + standardization
Role	Primary case study	Secondary validation

IV. PROPOSED Q-LEARNING HYPER-HEURISTIC

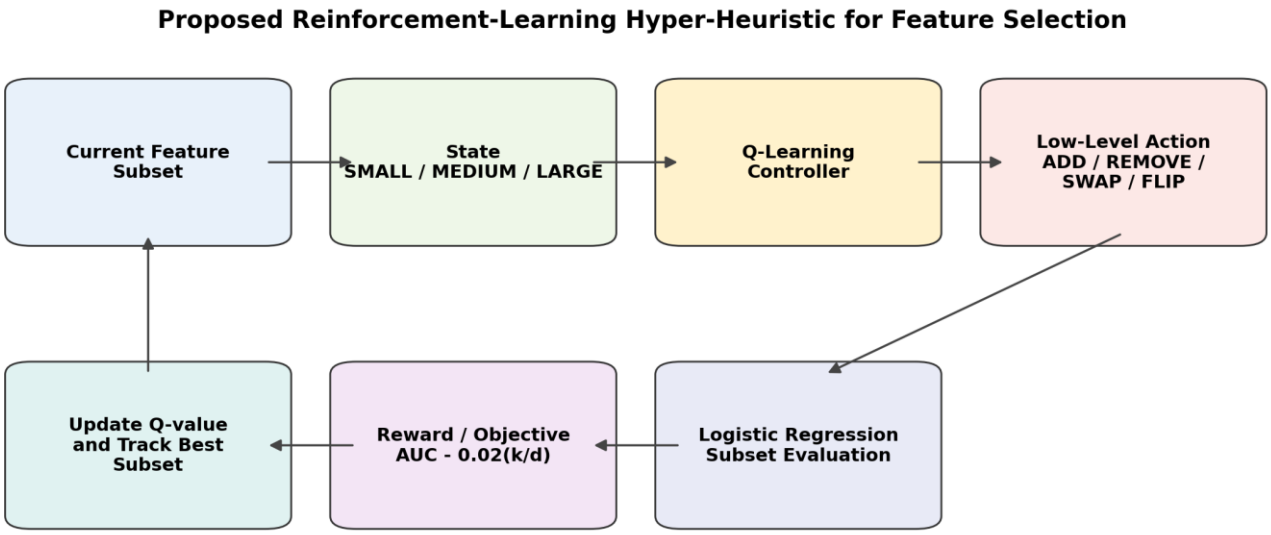


Figure 1. High-level architecture of the proposed Q-learning hyper-heuristic.

A. Low-Level Feature-Selection Heuristics

The high-level controller does not select individual features directly. Instead, each action invokes one of four low-level heuristics that modifies the current binary subset:

- ADD: choose one currently excluded feature and include it.
- REMOVE: remove one currently selected feature while preventing an empty subset.
- SWAP: remove one selected feature and replace it with one currently excluded feature.
- FLIP: toggle the inclusion state of one randomly selected feature.

The operators provide complementary search behaviors. ADD expands a subset, REMOVE promotes compactness, SWAP explores alternative compositions at constant subset size, and FLIP can either expand or contract the subset depending on the selected bit.

B. State Representation

To keep the Q-table compact, the search state is defined by the size of the current subset rather than by the full binary feature vector. The Diabetes controller uses SMALL for up to 7 selected features, MEDIUM for 8–14, and LARGE for 15–21. The Heart Disease controller uses SMALL for up to 4, MEDIUM for 5–8, and LARGE for 9–13. This abstraction reduces the state space to three states while still giving the controller information about whether the current solution is sparse, moderate, or large.

C. Q-Learning Policy

For state s and low-level heuristic a, the Q-value is updated according to the standard temporal-difference rule [4]:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

The implementation uses a learning rate $\alpha = 0.20$ and discount factor $\gamma = 0.80$. Actions are selected with an epsilon-greedy strategy. Exploration begins at $\varepsilon = 0.30$, decays by a factor of 0.95, and is bound below by 0.05. The immediate reward is the change in the optimization objective between the candidate and current subset, $r = J(z_new) - J(z_current)$.

D. Search and Final Evaluation

During search, candidate subsets are evaluated on development data, and repeated evaluations of the same subset are cached. The best subset encountered according to the optimization objective is retained. Final performance is then measured on held-out data that is not used for heuristic search. This separation is important because evaluating the selected subset on the same validation data used during search would provide an optimistically biased estimate.

Table 3. Core hyper-heuristic configuration.

Parameter	Diabetes	Heart Disease
RL states	3 (small/medium/large)	3 (small/medium/large)
Actions	ADD, REMOVE, SWAP, FLIP	ADD, REMOVE, SWAP, FLIP
Learning rate α	0.20	0.20
Discount factor γ	0.80	0.80
Initial ε	0.30	0.30
ε decay	0.95	0.95
Minimum ε	0.05	0.05
Main objective	AUC − 0.02(k/21)	AUC − 0.02(k/13)

V. EXPERIMENTAL METHODOLOGY

A. Predictive Model and Preprocessing

Logistic Regression is used as the predictive model for subset evaluation. Standardization is performed with StandardScaler. For Diabetes, class_weight="balanced" is used because of the target imbalance. For Heart Disease, median imputation is fitted on the training data before standardization. The implementation uses Python with NumPy, pandas, scikit-learn, SciPy, statsmodels, and Matplotlib. Scikit-learn provides the main preprocessing, classifier, and conventional feature-selection components [12].

B. Baseline Methods

RL hyper-heuristic is compared against several baselines designed to answer different questions. The all-feature model measures the cost of performing no selection. Random and fixed low-level-heuristic strategies test whether learning the operator choice adds value beyond applying operators without adaptation. SelectKBest provides a univariate filter baseline using the ANOVA F-statistic. L1-regularized Logistic Regression provides an embedded selector. RFE provides a wrapper-style recursive elimination baseline using Logistic Regression as the estimator.

C. Repeated Evaluation Design

The project contains exploratory single-split experiments and final repeated evaluations. The final Diabetes comparison uses 10 independent outer data splits. The final heart disease comparison uses 30 repeated splits because the dataset is much smaller and performance is more variable. Within each split, search/selection decisions are made using development data and final ROC-AUC is computed on the held-out test partition.

D. Metrics and Statistical Tests

ROC-AUC is the primary predictive-performance metric because it summarizes ranking performance across classification thresholds. The project also reports accuracy, precision, recall/sensitivity, F1-score, and specificity in baseline inspection stages. Feature-selection quality is summarized by the number of selected variables, percentage reduction relative to the full feature set, and the combined performance-compactness objective J(z).

For the main repeated method comparisons, paired Wilcoxon signed-rank tests are used because the same random splits/seeds provide naturally paired observations and the analysis does not require a normality assumption. Multiple comparisons are adjusted using the Holm sequentially rejective procedure [13]. Statistical significance is assessed at an adjusted $\alpha = 0.05$.

VI. RESULTS AND PERFORMANCE ANALYSIS

A. Initial Diabetes Baseline and Heuristic Experiments

Using all 21 Diabetes features on the initial held-out split produced ROC-AUC 0.8196, accuracy 0.7316, precision 0.3108, recall 0.7611, F1 0.4414, and specificity 0.7268. This established a strong reference point before feature-selection search. In early repeated RL experiments using a fixed split, the mean ROC-AUC was 0.8179 with 14.13 selected features and a mean feature reduction of 32.7%.

The fixed/random heuristic comparison showed that no one low-level operator dominated all others. ADD, SWAP, and RANDOM achieved mean AUC values around 0.8173–0.8176, while REMOVE produced more aggressive feature reduction but lower mean AUC. This provides a rationale for an

adaptive controller, although the later paired analysis showed that the learned RL policy was not significantly superior to RANDOM, ADD, or SWAP in the early fixed-split experiment.

Table 4. Exploratory 30-seed Diabetes heuristic comparison.

Method	Mean AUC	Mean Features	Reduction	Mean Objective
ADD	0.8176	14.53	30.8%	0.8038
FLIP	0.8156	12.73	39.4%	0.8035
RANDOM	0.8173	13.70	34.8%	0.8043
REMOVE	0.8120	11.67	44.4%	0.8009
SWAP	0.8175	13.83	34.1%	0.8043
RL	0.8179	14.13	32.7%	—

B. Final Diabetes Comparison

Table 5. Final Diabetes repeated evaluation across 10 splits.

Method	Mean AUC	Std AUC	Mean Features	Reduction	Mean Objective
All Features	0.8230	0.0014	21.00	0.0%	0.8030
L1	0.8224	0.0013	11.00	47.6%	0.8119
RFE	0.8221	0.0016	9.80	53.3%	0.8128
RL	0.8210	0.0016	13.80	34.3%	0.8079
SelectKBest	0.8178	0.0015	8.00	61.9%	0.8102

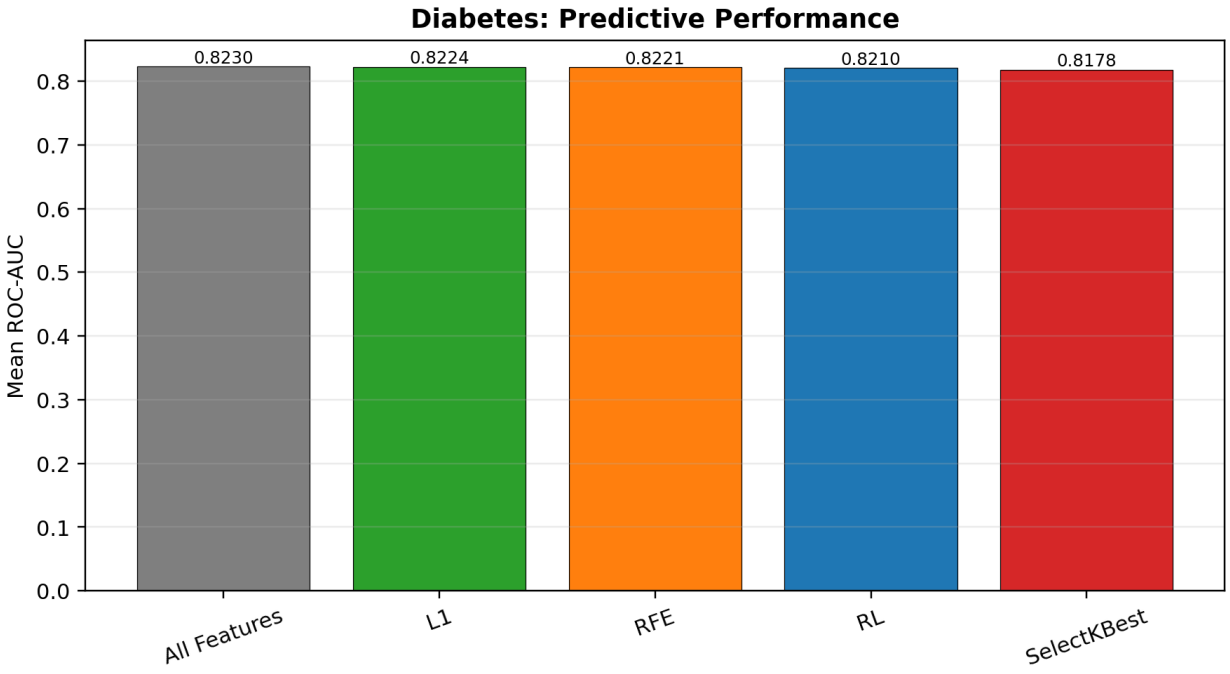


Figure 2. Mean ROC-AUC of the final Diabetes methods.

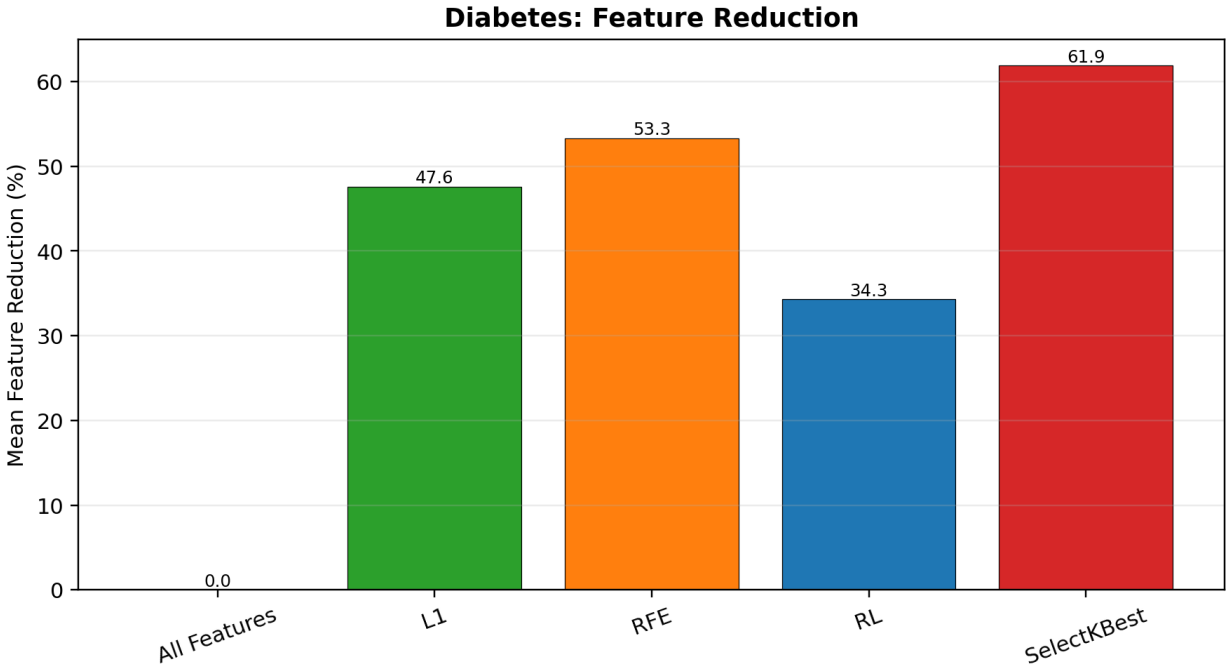


Figure 3. Mean feature reduction of the final Diabetes methods.

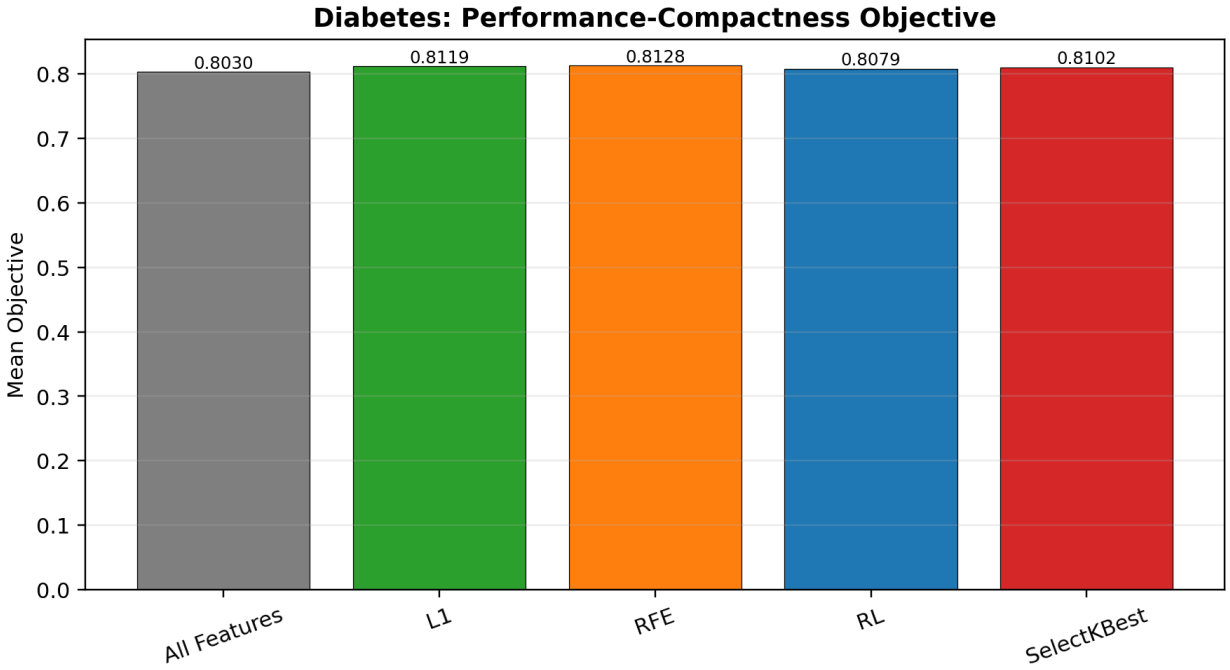


Figure 4. Mean performance-compactness objective on Diabetes.

The full-feature model achieved the highest mean ROC-AUC (0.8230), while the RL method achieved 0.8210 using 13.8 features on average. The absolute mean AUC difference between RL and all features was only −0.001924, but RL removed 7.2 features on average and improved the mean optimization objective from 0.8030 to 0.8079. This indicates that RL preserved most of the predictive discrimination while reducing roughly one-third of the variables.

The conventional methods were nevertheless strong. RFE achieved the highest mean objective (0.8128) with 9.8 features on average, and L1 achieved 0.8119 with 11 features. SelectKBest reduced the feature set most aggressively to 8 variables but also had the lowest mean AUC (0.8178). Thus, RL provided a useful compromise but was not the best method under the selected objective.

Table 6. Final Diabetes paired Wilcoxon results after Holm correction (RL minus comparator).

Comparison	Δ AUC	Holm p (AUC)	Δ Features	Holm p (Feat.)	Δ Objective	Holm p (Obj.)
RL vs All Features	−0.001924	0.007812	−7.20	0.007812	+0.004933	0.007812
RL vs L1	−0.001373	0.007812	+2.80	0.007812	−0.004039	0.007812
RL vs RFE	−0.001104	0.019531	+4.00	0.007812	−0.004914	0.007812
RL vs SelectKBest	+0.003235	0.007812	+5.80	0.007812	−0.002288	0.007812

After Holm correction, RL had significantly lower AUC than the all-feature, L1, and RFE models, although the numerical differences were small. RL had significantly higher AUC than SelectKBest. For the combined objective, RL significantly improved on all features but was significantly worse than L1, RFE, and SelectKBest. Therefore, the final statistical analysis supports a nuanced conclusion: RL is effective at obtaining compact high-performing subsets, but established methods—especially RFE and L1—provide stronger performance-compactness trade-offs on the large Diabetes dataset.

C. Final Heart Disease Comparison

Table 7. Final Heart Disease repeated evaluation across 30 splits.

Method	Mean AUC	Std AUC	Mean Features	Reduction	Mean Objective
All Features	0.9032	0.0268	13.00	0.0%	0.8832
L1	0.8926	0.0281	8.00	38.5%	0.8803
RFE	0.8911	0.0314	9.07	30.3%	0.8772
RL	0.8791	0.0373	8.03	38.2%	0.8667
SelectKBest	0.8905	0.0315	7.73	40.5%	0.8786

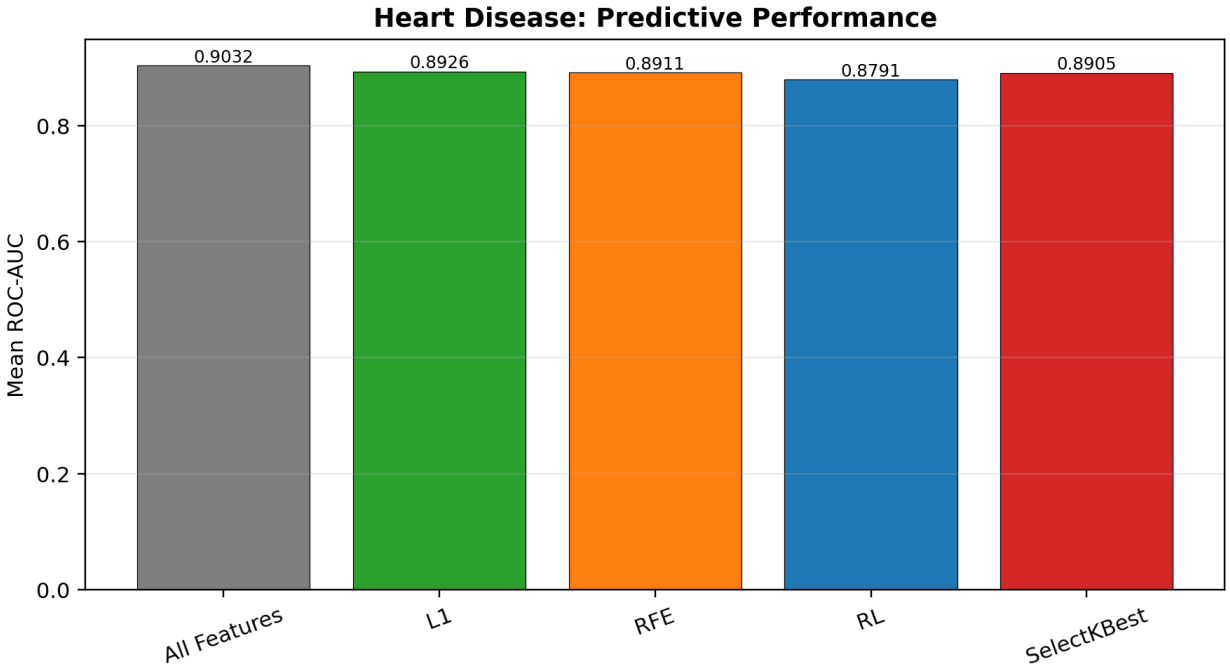


Figure 5. Mean ROC-AUC of the final heart disease methods.

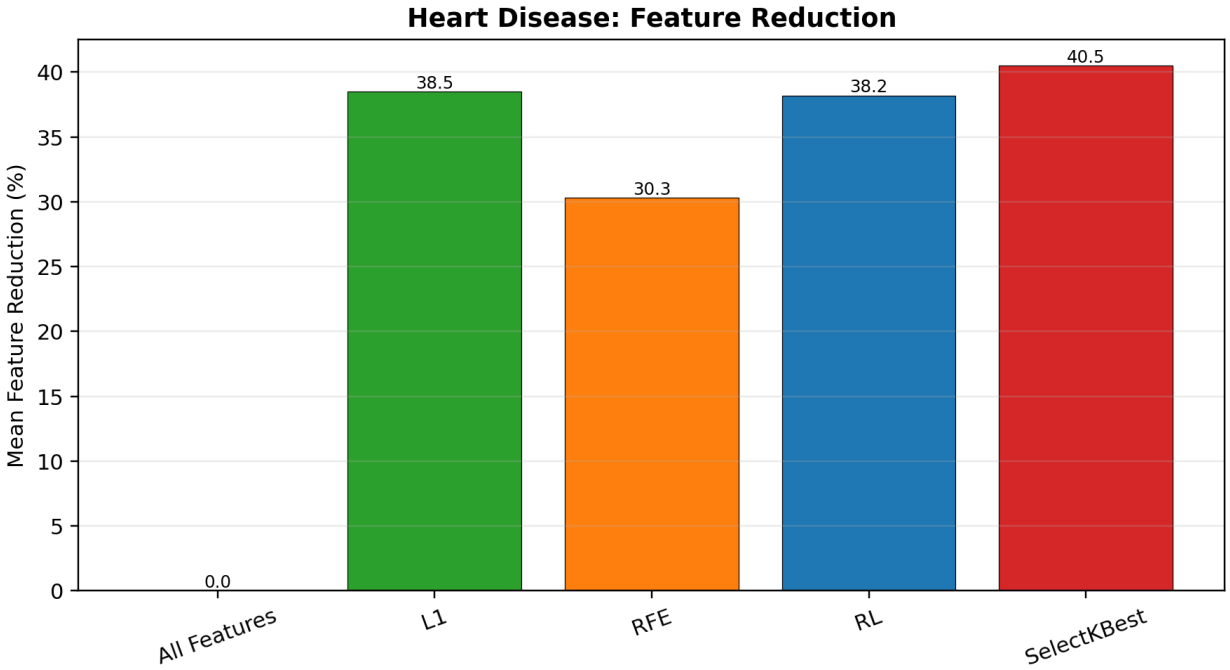


Figure 6. Mean feature reduction of the final heart disease methods.

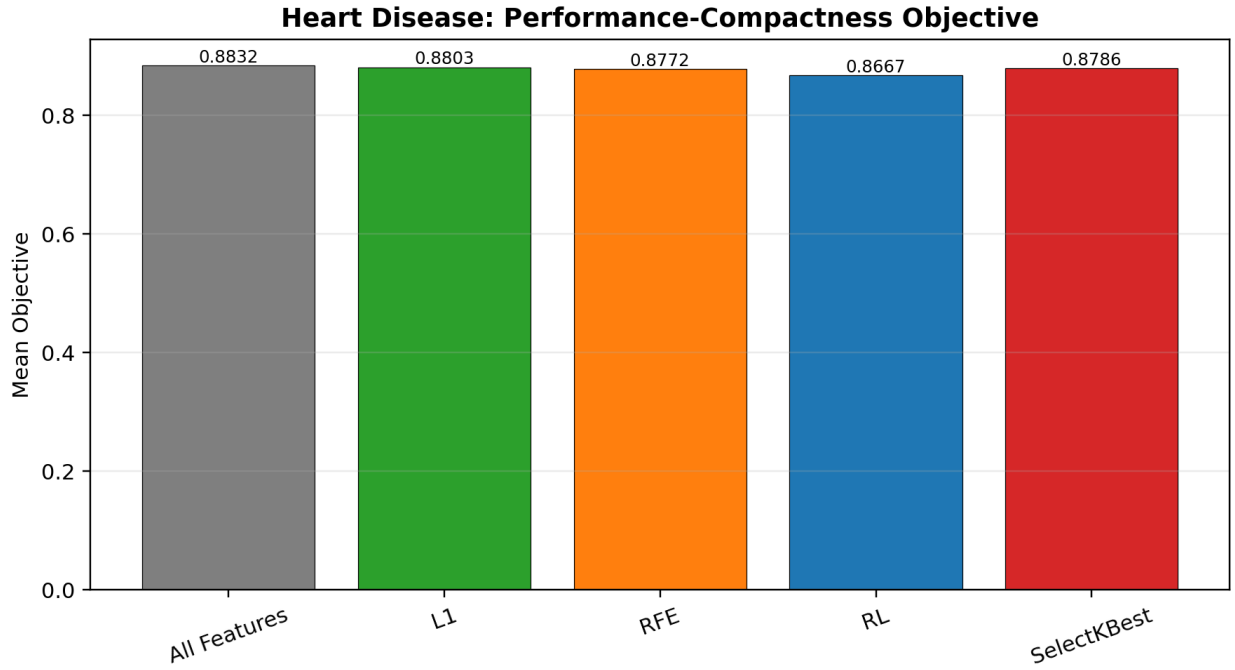


Figure 7. Mean performance-compactness objective on heart disease.

The Heart Disease results differ from the Diabetes results. The all-feature model achieved the best mean AUC (0.9032). RL selected approximately 8.03 of 13 features, corresponding to a 38.2% reduction, but its mean AUC fell to 0.8791 and had a larger standard deviation (0.0373). Across the 30 repeated runs, RL exceeded the all-feature AUC only 4 times and lost 26 times, yielding a mean AUC difference of approximately −0.0241.

The conventional feature selectors also reduced the feature set but retained more predictive performance. L1 achieved 0.8926, RFE 0.8911, and SelectKBest 0.8905. The larger run-to-run variability indicates that feature-subset evaluation is substantially noisier in this 303-observation dataset than in the 253,680-observation Diabetes dataset.

Table 8. Final Heart Disease paired Wilcoxon results after Holm correction (RL minus comparator).

Comparison	Δ AUC	Holm p (AUC)	Δ Features	Holm p (Feat.)	Δ Objective	Holm p (Obj.)
RL vs All Features	−0.024098	0.000090	−4.97	0.000006	−0.016457	0.016127
RL vs L1	−0.013546	0.047719	+0.03	1.000000	−0.013597	0.033231
RL vs RFE	−0.012031	0.168020	−1.03	0.345629	−0.010441	0.140283
RL vs SelectKBest	−0.011382	0.168020	+0.30	1.000000	−0.011843	0.049572

RL was significantly worse than the all-feature model in AUC and objective while selecting significantly fewer variables. Relative to L1, RL also had significantly lower AUC and objective with essentially the same feature count. Differences from RFE were not significant after Holm correction. Against SelectKBest, AUC and feature-count differences were not significant, but the RL objective was significantly lower. These results do not support an overall superiority claim for RL on the small heart disease dataset.

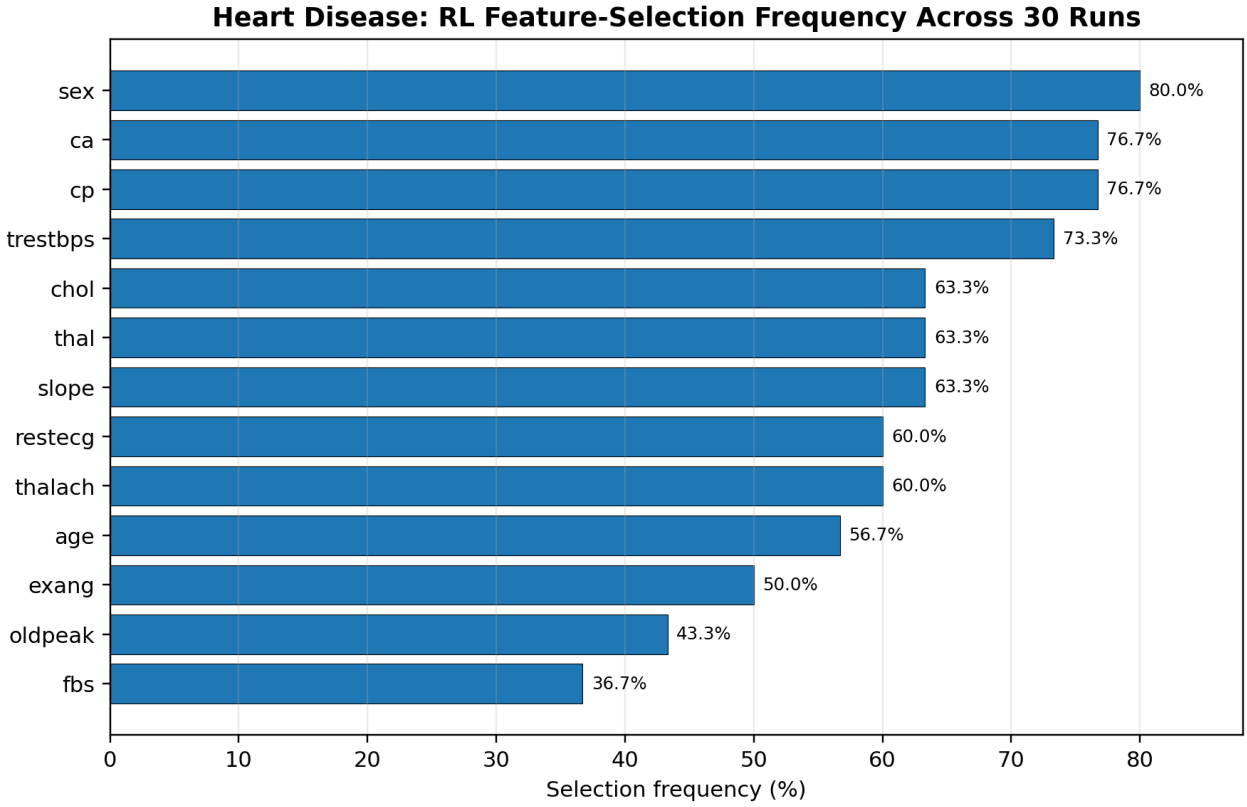


Figure 8. Frequency with which each heart disease variable was selected by RL across 30 repeated experiments.

Feature-selection frequency provides additional information about RL search behavior. The variables sex, ca, and cp were selected in approximately 80.0%, 76.7%, and 76.7% of the repeated experiments, respectively, followed by trestbps at 73.3%. The pattern is not uniform, suggesting that the search repeatedly identifies some variables as useful. However, frequent selection does not by itself establish clinical importance or causal relevance; it only reflects behavior under this model, objective, and validation procedure.

D. Cross-Dataset Analysis

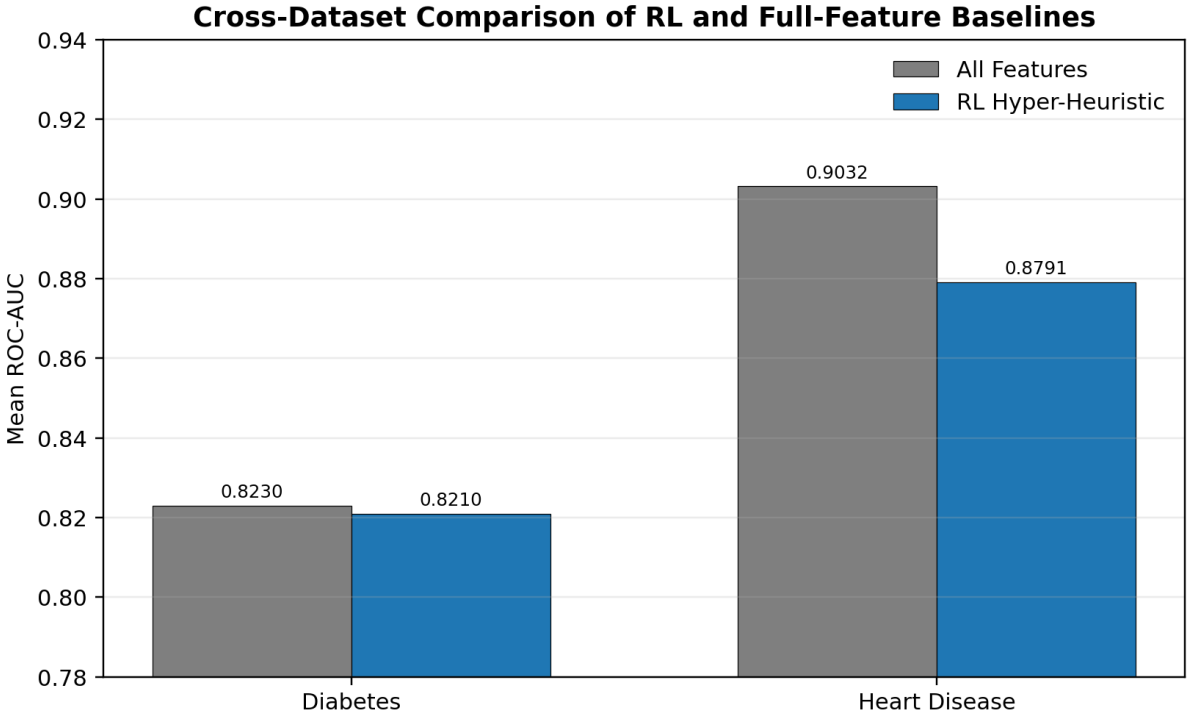


Figure 9. Cross-dataset mean ROC-AUC for RL and the full-feature baseline.

The two datasets expose different operating regimes for the same search concept. On Diabetes, the RL mean AUC is only 0.0019 below the full-feature model while reducing the feature set by 34.3%. On Heart Disease, the mean difference expands to about 0.0241 while the standard deviation rises substantially. This contrast is consistent with greater validation noise and subset instability when only a few hundred observations are available. It should not be interpreted as proof that dataset size alone causes the difference, because the datasets also differ in features, target characteristics, and preprocessing.

VII. DISCUSSION

A. What the Results Support

The principal finding is that Q-learning can successfully act as a high-level feature-selection controller. The method navigates the subset search space through learned heuristic choices and repeatedly discovers substantially smaller feature sets. On the large Diabetes dataset, it preserves ROC-AUC close to the full model and improves the chosen performance-compactness objective relative to using all variables. This validates the core feasibility claim in the project proposal.

At the same time, the experiments show why a research-oriented comparison must include strong non-RL baselines. RFE and L1 achieve better mean objectives on Diabetes, and the all-feature model remains strongest in raw AUC. On Heart Disease, RL is less stable and is generally weaker than the full-feature and L1 approaches. The appropriate conclusion is therefore not that RL replaces conventional feature selection, but that adaptive hyper-heuristics provide an additional search strategy whose usefulness is problem dependent.

B. Interpretation of the Adaptive Hyper-Heuristic

A key conceptual difference between this method and direct RL feature selection is the level at which learning occurs. The Q-learning agent does not assign one action to each individual feature. Instead, it chooses a search operator based on the coarse subset-size state. This makes the controller compact and easy to interpret: the learned policy concerns how the search should move, while the low-level heuristics determine which feature is modified. The approach therefore follows the hyper-heuristic principle of searching over heuristics rather than directly over complete solutions [3].

The simplicity of the state representation is both strength and a limitation. Three states keep learning inexpensive, but they ignore which specific variables are selected and how recently the search has improved. Two subsets with the same cardinality can have very different predictive quality, yet they are represented by the same state. A richer state description may allow the controller to distinguish search contexts more effectively.

C. Healthcare Interpretation and Scope

The models in this project predict dataset labels from available health indicators; they are not clinical diagnostic systems. Selected variables should not be interpreted as causal risk factors or clinical recommendations. Feature-selection frequency reflects the behavior of the optimization procedure and classifier under repeated splits, not medical importance. Any clinical use would require domain validation, external cohorts, calibration analysis, fairness assessment, and prospective evaluation.

D. Limitations

- The RL state contains only subset-size information; it does not encode feature identity, recent improvement, or model uncertainty.
- The low-level heuristics modify features randomly within each operator, so the controller learns which type of move to use but not which individual feature to add or remove.
- Logistic Regression is the only predictive model used for the main wrapper evaluation; performance may differ with tree ensembles, boosting, or nonlinear models.
- The compactness penalty $\lambda = 0.02$ is fixed. Different application costs could justify different trade-offs between AUC and feature count.
- The Heart Disease dataset is small, and repeated validation therefore produces substantial variability. The observed cross-dataset difference cannot be attributed to sample size alone.
- No external clinical chores are used. The secondary Heart dataset is a second benchmark, not a clinical external-validation study.

- The project optimizes statistical prediction and subset size; it does not model feature acquisition cost, patient burden, fairness, or missingness at deployment.

E. Future Work

Several extensions follow naturally from these limitations. A richer RL state could include recent objective improvement, validation uncertainty, and subset diversity. The action layer could be hierarchical, with the high-level agent first choosing ADD/REMOVE/SWAP/FLIP and a second policy choosing the specific feature. Multi-objective RL could model AUC, feature count, measurement cost, and fairness explicitly rather than combining them into a single fixed-weight reward. Repeated nested cross-validation and additional healthcare datasets would strengthen generalization claims. Finally, explainability methods could be used to compare the stability of selected features with coefficient-based or SHAP-based explanations.

VIII. CONCLUSION

This research implemented and evaluated a reinforcement-learning-based hyper-heuristic for feature selection in disease prediction. Q-learning was used as a high-level controller to select among ADD, REMOVE, SWAP, and FLIP operations, while Logistic Regression evaluated the predictive quality of the resulting subsets. The method was tested on the CDC Diabetes Health Indicators dataset and the UCI Cleveland Heart Disease dataset and compared with full-feature models, random/fixed heuristics, SelectKBest, L1 selection, and RFE.

On Diabetes, the RL method achieved a mean ROC-AUC of 0.8210 while reducing the 21-feature set by 34.3%, demonstrating that adaptive heuristic selection can obtain compact subsets with only a small loss in discrimination. However, RFE and L1 achieved stronger performance-compactness objectives. On Heart Disease, RL reduced the feature set by 38.2% but showed lower and more variable ROC-AUC, and it did not outperform the strongest conventional baselines. Paired Wilcoxon tests with Holm correction confirmed that several of these differences were statistically meaningful.

The overall conclusion is therefore balanced: reinforcement-learning hyper-heuristics are a viable and interpretable approach for adaptive feature-selection search, but they are not universally superior to established methods. Their effectiveness depends on the stability of subset evaluation, dataset characteristics, and the design of the state, reward, and low-level heuristics. The project demonstrates the value of evaluating emerging AI optimization techniques against strong baselines and reporting both successful and unfavorable results rather than assuming that a more complex adaptive method must perform best.

IX. REPRODUCIBILITY AND PROJECT ORGANIZATION

The implementation is organized as a reproducible Python project. Raw dataset archives are stored under 01_data, source scripts under 02_scripts, experimental CSV files under 03_results, and publication figures under 04_figures. Paths are centralized in project_paths.py. The requirements file lists NumPy, pandas, scikit-learn, SciPy, statsmodels, and Matplotlib. The final report and presentation directories are reserved under 05_report and 06_presentation. All Python scripts were syntax-checked after path refactoring, and the dataset/result directories were verified to exist.

Table 9. Mapping of the final project artifacts to the research workflow.

Stage	Representative scripts / artifacts
Data inspection / baselines	01–05; 15–17
RL hyper-heuristic prototypes	06–08; 18
Repeated heuristic comparisons	09, 11–14
Heart repeated RL/statistics	19–20
Final conventional baselines	21, 24
Final repeated comparisons/statistics	22–23, 25
Consolidation and figures	26_final_results_consolidation.py
Outputs	03_results/* and 04_figures/*

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APPENDIX A. FINAL NUMERICAL SUMMARY

For transparency, the key numerical results used in the final interpretation are consolidated below. These are the values generated by the final repeated evaluation and statistical-comparison scripts.

Table 10. Consolidated cross-dataset RL summary.

Dataset	RL Mean AUC	Std AUC	Mean Features	Reduction	Mean Objective	All-Feature AUC
Diabetes	0.8210	0.0016	13.80	34.3%	0.8079	0.8230
Heart Disease	0.8791	0.0373	8.03	38.2%	0.8667	0.9032